

Step-by-step guide: Reconstructing Individual Patient Data (IPD) from Digitized KM Curves

Using **IPDfromKM** with multi-endpoint Excel inputs (4 sheets) and best-practice validation, QC, and saving.

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1. Purpose and scope

This document describes a reproducible workflow to reconstruct time-to-event individual patient data (IPD) from published Kaplan-Meier (KM) curves that have been digitized into an Excel file. The workflow is designed to handle multiple endpoints, each stored on a separate Excel sheet, and to generate a separate, validated IPD dataset (RDS) per endpoint.

The approach follows the standard IPD-from-KM pipeline: (i) verify digitized curve points, (ii) clean and enforce KM properties, (iii) reconstruct IPD using risk tables, (iv) validate the reconstruction via diagnostics and overlays, and (v) save endpoint-specific outputs.

2. Inputs and required structure

2.1 Excel file layout

Sheet	Endpoint name (used in code)	Description / source
1	MD_RSV_LRTI	Medically attended RSV-LRTI (Simões 2023)
2	RSV_Hospitalization	RSV hospitalization (Simões 2023)
3	Severe_RSV_LRTI	Severe RSV-LRTI (Kampmann 2023)
4	Any_RSV_LRTI	RSV-LRTI (any) (Kampmann 2023)

2.2 Required columns per sheet

Each sheet must contain three columns representing: time, KM survival proportion, and treatment arm. Your code auto-detects typical names, but the simplest is to standardize column headers to:

- **time:** numeric time since randomization/dose (days).
- **proportion:** numeric survival probability $S(t)$ on $[0,1]$ (may have small digitization noise).
- **treat:** arm label (e.g., placebo, nirsevimab).

If other names are used (e.g., Days, Survival, Arm), either rename them in Excel or extend the standardization function to map columns explicitly per endpoint.

2.3 Risk tables

IPD reconstruction requires numbers at risk at a set of landmark times (the paper's risk table). For each endpoint, you must supply trisk and two vectors of nrisk (one per arm) with equal length.

3. Step-by-step workflow

The workflow below assumes all four sheets are populated.

Step 0 - Set up the workspace

```
suppressPackageStartupMessages({
  library(readxl)
  library(dplyr)
  library(tidyr)
  library(stringr)
  library(ggplot2)
  library(purrr)
  library(glue)
  library(IPDfromKM)
  library(survival)
})

set.seed(42)
options(stringsAsFactors = FALSE, dplyr.summarise.inform = FALSE)
```

Step 1 - Configure endpoints and file paths

```
excel_path <- "data/extracted_points.xlsx"

out_dir <- "data/reconstructed_ipd"
dir.create(out_dir, recursive = TRUE, showWarnings = FALSE)

endpoints <- tibble::tibble(
  sheet = 1:4,
  endpoint = c("MD_RSV_LRTI", "RSV_Hospitalization", "Severe_RSV_LRTI", "Any_RSV_LRTI")
)

t_max <- 150
trisk <- c(0, 30, 60, 90, 120, 150)

risk_tables <- list(
  MD_RSV_LRTI = list(
    placebo = c(786, 772, 756, 737, 729, 724),
    nirsevimab = c(1564, 1553, 1546, 1538, 1527, 1519)
  ),
  RSV_Hospitalization = list(
    placebo = c(786, 778, 769, 761, 757, 753),
    nirsevimab = c(1564, 1554, 1547, 1540, 1535, 1529)
  ),
  Severe_RSV_LRTI = list(placebo = c(...), nirsevimab = c(...)),
  Any_RSV_LRTI = list(placebo = c(...), nirsevimab = c(...))
)

arm_map <- c("placebo" = "placebo", "nirsevimab" = "nirsevimab")
```

Step 2 - Implement utilities and validation checks

- **stop_if()** to fail fast on structural problems (missing arms, invalid risk tables).
- **inform()** for consistent logging (glue evaluated in caller environment).
- **standardize_km_columns()** to normalize sheet columns to time/surv/treat.
- **clean_km_arm()** to enforce KM properties (origin, duplicates, bounds, monotonicity).

- **km_critical_checks()** to flag digitization issues early (including small survival overshoots).

```
stop_if <- function(cond, msg) if (isTRUE(cond)) stop(msg, call. = FALSE)
inform <- function(...) message(glue::glue(..., .envir = parent.frame()))
```

```
clean_km_arm <- function(df_arm) {
  df_arm %>%
    filter(!is.na(time), !is.na(surv)) %>%
    mutate(time = as.numeric(time), surv = as.numeric(surv)) %>%
    bind_rows(tibble::tibble(time = 0, surv = 1)) %>%
    arrange(time) %>%
    group_by(time) %>%
    summarise(surv = min(surv), .groups = "drop") %>%
    mutate(
      surv = pmin(pmax(surv, 0), 1),
      surv = cummin(surv)
    )
}
```

Step 3 - Reconstruct IPD for one endpoint

```
reconstruct_one_endpoint <- function(excel_path, sheet, endpoint,
                                     t_max, trisk, risk_tables, arm_map) {

  raw <- read_excel(excel_path, sheet = sheet) %>%
    standardize_km_columns() %>%
    mutate(
      treat = str_to_lower(str_trim(treat)),
      treat = recode(treat, !!!arm_map)
    )

  mono_raw <- km_critical_checks(raw, endpoint, t_max)

  km_p <- raw %>% filter(treat == "placebo") %>% clean_km_arm() %>% ensure_tmax(t_max)
  km_n <- raw %>% filter(treat == "nirsevimab") %>% clean_km_arm() %>% ensure_tmax(t_max)

  risk_entry <- risk_tables[[endpoint]]
  validate_risk_table(endpoint, trisk, risk_entry)

  pre_p <- preprocess(dat = km_p, trisk = trisk, nrisk = risk_entry$placebo, maxy = 1)
  pre_n <- preprocess(dat = km_n, trisk = trisk, nrisk = risk_entry$nirsevimab, maxy = 1)

  est_p <- getIPD(pre = pre_p, armID = 0)
  est_n <- getIPD(pre = pre_n, armID = 1)

  ipd <- bind_rows(
    est_p$IPD %>% mutate(treat = "placebo"),
    est_n$IPD %>% mutate(treat = "nirsevimab")
  ) %>%
  mutate(
    treat = factor(treat, levels = c("placebo", "nirsevimab")),
    time = as.numeric(time),
    status = as.integer(status),
    endpoint = endpoint
  )

  diagnostics <- list(
    precision = list(placebo = est_p$precision, nirsevimab = est_n$precision),
    kstest = list(placebo = est_p$kstest, nirsevimab = est_n$kstest),
    n_by_arm = table(ipd$treat),
    events_by_arm = table(ipd$treat, ipd$status)
  )

  list(bundle = list(endpoint = endpoint, sheet = sheet, t_max = t_max,
                    trisk = trisk, nrisk = risk_entry,
                    km_clean = list(placebo = km_p, nirsevimab = km_n),
                    ipd = ipd, diagnostics = diagnostics))
}
```

Step 4 - Run reconstruction for all endpoints and save RDS

```
results <- endpoints %>%
  mutate(out_rds = file.path(out_dir, paste0(endpoint, ".rds"))) %>%
  pmap(function(sheet, endpoint, out_rds) {

    res <- reconstruct_one_endpoint(
      excel_path = excel_path,
      sheet = sheet,
      endpoint = endpoint,
```

```

      t_max      = t_max,
      trisk      = trisk,
      risk_tables = risk_tables,
      arm_map    = arm_map
    )

    saveRDS(res$bundle, out_rds)
    inform("Saved: {out_rds}")

    list(endpoint = endpoint, out_rds = out_rds)
  })

```

Step 5 - Quality control and validation

- Check that N by arm equals the baseline risk table at time 0.
- Check event counts by arm and compare to the paper (if reported).
- Review IPDfromKM diagnostics (precision and KS tests).
- Overlay reconstructed KM vs digitized points for each arm.
- Save a reconstruction bundle (inputs + outputs) for auditability.

```

bundle <- readRDS(file.path(out_dir, "MD_RSV_LRTI.rds"))
ipd <- bundle$ipd

fit <- survfit(Surv(time, status) ~ treat, data = ipd)
plot(fit)

cox <- coxph(Surv(time, status) ~ treat, data = ipd)
summary(cox)

```

4. Common issues and fixes

Column-name mismatch across sheets

If `standardize_km_columns()` cannot identify time/surv/treat uniquely, standardize headers in Excel or implement an explicit column map per endpoint.

Survival outside [0,1] and non-monotone curves

Small overshoots and tiny increases are common digitization artifacts. The recommended approach is to log them, clamp survival to [0,1], and enforce non-increasing curves with `cummin()`. Large problems indicate extraction errors and should be corrected upstream.

Risk table inconsistencies

Risk tables must be aligned with trisk and non-increasing. Stop and correct if lengths differ or values increase over time.

5. Output specification

Each endpoint RDS contains: endpoint, sheet, `t_max`, trisk, nrisk (per arm), cleaned KM points, reconstructed IPD, and reconstruction diagnostics. This supports downstream Cox models, flexible hazards, and vaccine efficacy waning analyses.